10 Show that $(x + 1)(x + 2)(x + 3)$ can be written in the form $ax^3 + bx^2 + cx + d$ where a, b, c and d are positive integers.

First multiply $(x + 1)(x + 2)$.

×	x	$+1$
x	x^2	$+x$
$+2$	$+2x$	$+2$

$= x^2 + 3x + 2$

1 mark
Multiply two brackets

Now multiply by $(x + 3)$.

×	x^2	$+3x$	$+2$
x	x^3	$+3x^2$	$+2x$
$+3$	$+3x^2$	$+9x$	$+6$

1 mark
Multiply the quadratic by the third bracket

$x^3 + 6x^2 + 11x + 6$

1 mark
Collect all like terms

$a = 1$, $b = 6$, $c = 11$, $d = 6$

(Total for Question 10 is 3 marks)